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Introduction to Information and Data Security



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Information security is the discipline of protecting data, systems, and networks from unauthorized access, disclosure, alteration, or destruction.



Protects Data

Digital and physical information assets



Secures Systems

Hardware, software, and networks



Preserves Trust

Confidence of users and stakeholders

Information Assets, Security Challenges in Data-Driven Organizations



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Data Assets

Records, databases, IP, credentials



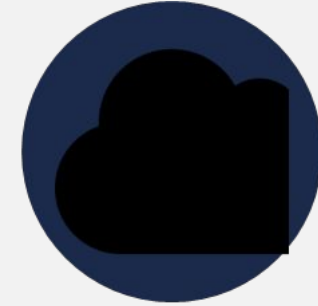
Infrastructure

Servers, networks, endpoints



People

Employees, vendors, third parties



Cloud & Services

SaaS platforms, storage, APIs

Key Challenges: exploding data volume, distributed cloud environments, rising insider & external threats, and continuous regulatory compliance pressure (NAAC/ISO/IT Act).

Importance and Objectives of Information Security



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Protect Assets

Safeguard data, systems, reputation



Build Trust

Assure stakeholders & customers



Ensure Compliance

Meet legal & regulatory mandates



Business Continuity

Minimize disruption & downtime

Confidentiality, Integrity and Availability (CIA Triad)



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Confidentiality



**CIA
TRIAD**

Confidentiality

Ensures information is disclosed only to authorized individuals — enforced via encryption and access control.

Integrity

Guarantees data accuracy and completeness — protected via checksums, hashing, and version control.

Availability

Ensures timely, reliable access to information — supported by redundancy, backups, and DR planning.



Availability

Real-World Applications of the CIA Triad



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Banking & Finance

Confidentiality

Encrypted transactions, masked account data



Healthcare

Integrity

Accurate, tamper-proof patient records (HIPAA)



Cloud Services

Availability

Redundant servers ensure 99.9% uptime SLAs

Threats, Vulnerabilities, Risks and Attack Surface



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A threat exploits a vulnerability, creating risk; the sum of all exploitable entry points forms the attack surface.

Data Breaches and Fundamentals of Risk Assessment



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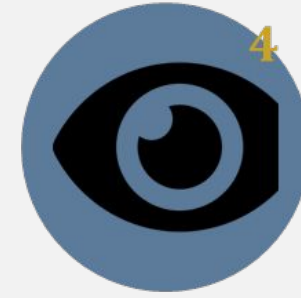
1



2



3



4

Identify

Assess

Evaluate likelihood and impact

Mitigate

Apply controls to reduce risk

Monitor

Continuously review & respond

A data breach is a confirmed incident where sensitive information is accessed or disclosed without authorization — often the trigger event that risk assessment aims to prevent.

Social Engineering, Phishing and Human-Based Attacks



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Phishing

Fraudulent emails/messages tricking users into revealing data



Pretexting

Fabricated scenario to extract information



Baiting

Enticing offers/media used to lure victims

Human-based attacks exploit psychology — trust, urgency, and fear — rather than technical flaws.

Identity Theft and Insider Threats



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Identity Theft

- Stolen personal/credential data
- Used for fraud or impersonation
- Often via phishing or data leaks



Insider Threats

- Originates from employees or contractors
- Can be malicious or negligent
- Hardest to detect — trusted access

Authentication, Authorization and Access Control



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Identify

Authenticate

Authorize

Access

Who are you? (username)

Prove identity
(password/biometric)

Determine permitted actions

Grant entry to resource

Access Control Models: Discretionary (DAC), Mandatory (MAC), and Role-Based (RBAC) — each governs how authorization decisions are enforced.

Password Security and Multi-Factor Authentication (MFA)



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Something You Know + Have + Are = Strong Authentication



Knowledge

Password / PIN

Something you know



Possession

OTP / Security token

Something you have



Inherence

Biometrics

Something you are

Need for Cryptography and Data Encryption



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Plaintext



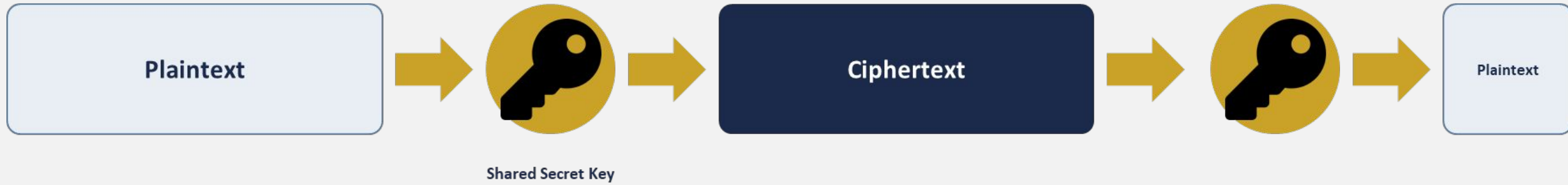
Ciphertext

- Protects sensitive data in storage and transit
- Ensures confidentiality even if data is intercepted
- Supports integrity and authenticity verification
- Mandated by compliance frameworks (IT Act, GDPR)

Symmetric Encryption and Its Applications



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Same key encrypts and decrypts the message — fast, efficient, and ideal for large data volumes.

AES for disk & file encryption

TLS bulk data transfer

Database & VPN encryption

Asymmetric Encryption and Its Applications



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A mathematically linked key pair — data encrypted with the public key can only be decrypted with the matching private key.

Secure key exchange (TLS/SSL)

Digital signatures & certificates

Email encryption (PGP/S-MIME)

Digital Signatures and Cryptographic Applications



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Document

Original message/file



Hash

Fixed-length digest generated



Sign

Hash encrypted with sender's
private key



Verify

Receiver validates using public
key

Digital signatures provide authenticity, integrity, and non-repudiation — core to e-governance, IT Act 2000 compliance, and secure digital transactions.

Thank you